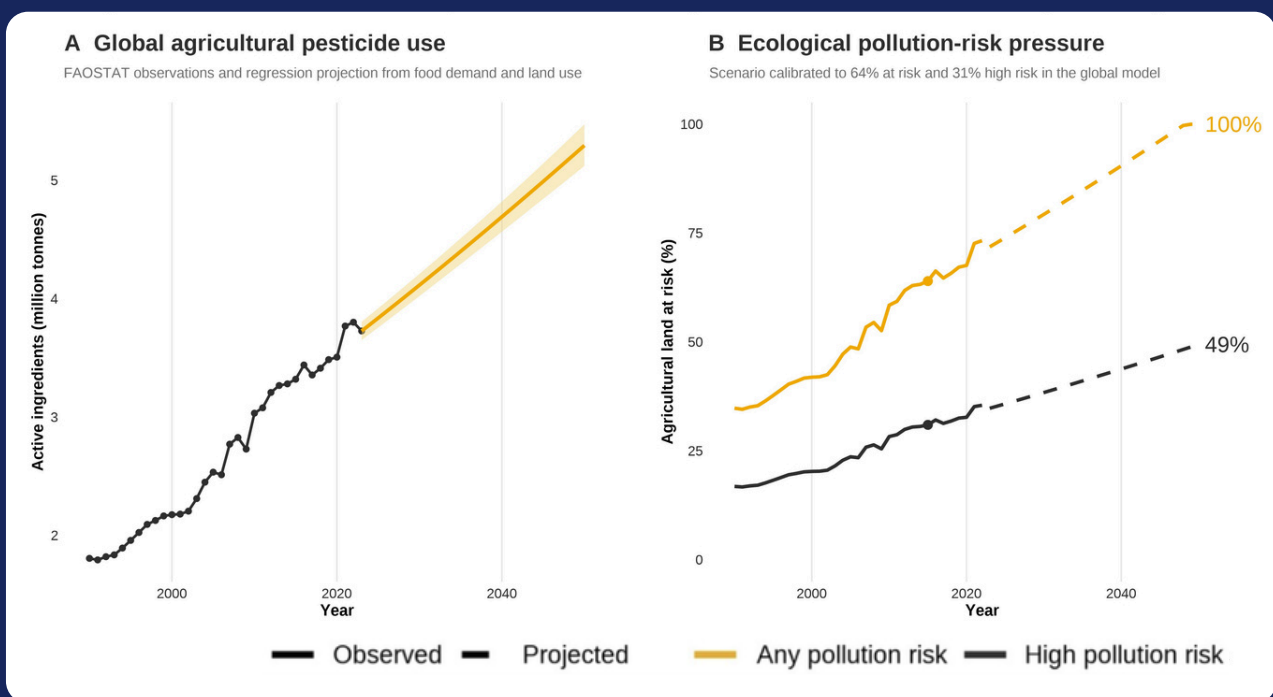




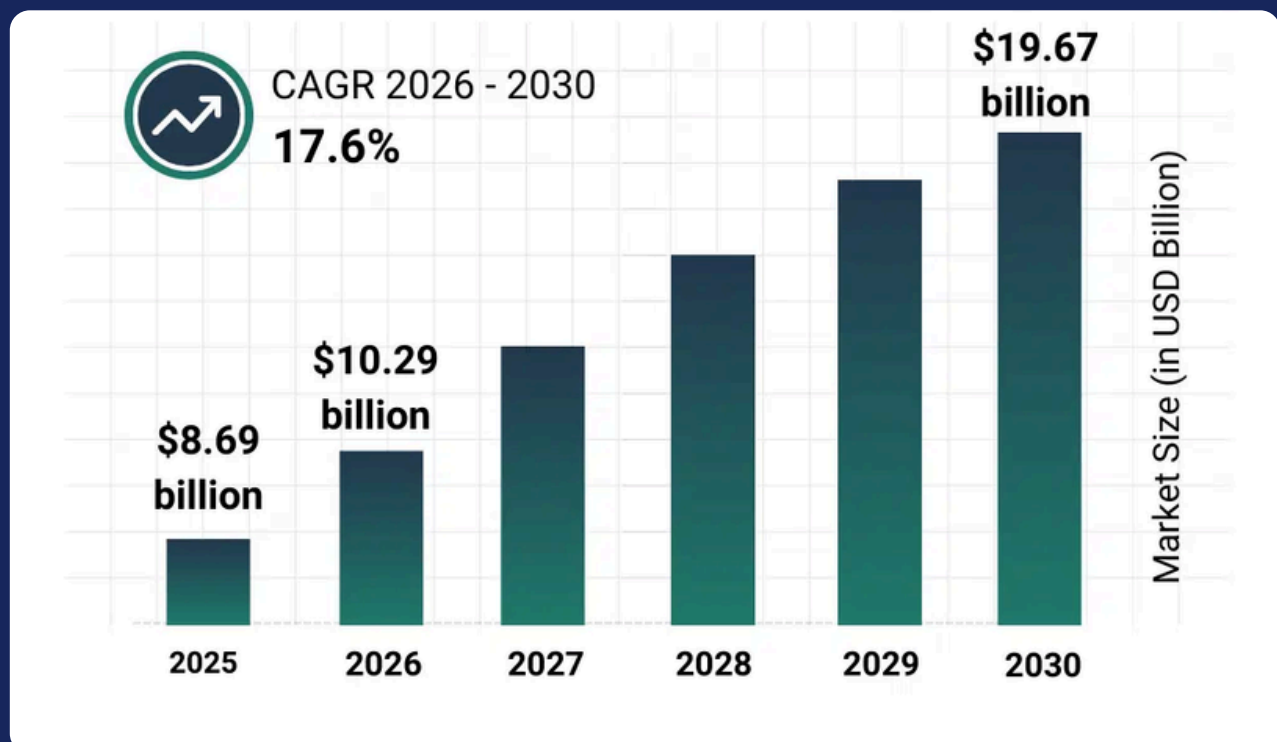
The Problem

By 2050, agriculture must produce 50% more food, yet pests already destroy up to 40% of crops and cause more than **\$220 billion in annual losses**. Climate change is expanding pest ranges, while chemical controls drive resistance, toxic residue accumulation and severe ecological damage through non-target effects.



Projected estimates using data from FAOSTAT show that pesticide use will increase by 25% over the next 25 years. These estimates suggest **by 2050, 100% of agricultural land will be at risk of pesticide based pollution**, with almost 50% of agricultural land being at high risk. This has generated an urgent need for more sustainable alternatives to avoid the immeasurable strain on global economies and biodiversity.

RNA interference is an effective, sustainable alternative to industrially produced chemicals. It can silence essential pest genes with sequence-level precision, minimising harmful effects of pesticide pollution and cleanup. The biopesticide market is predicted to expand over the coming years; however, widespread adoption of RNA as a biopesticide is impeded by the six key factors.

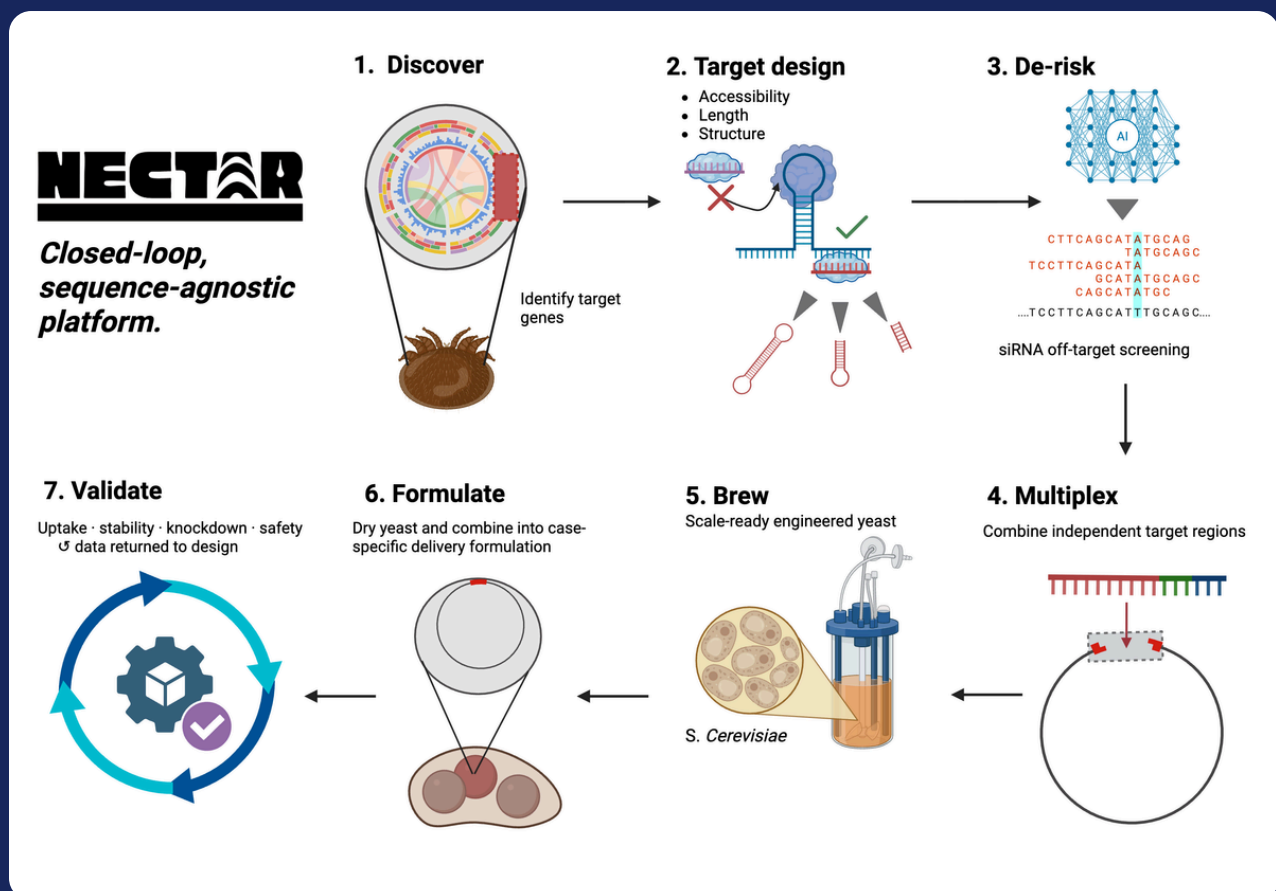


Factors limiting RNA biopesticide use:

- 1 RATIONAL TARGET SELECTION
- 2 OFF-TARGET SCREENING
- 3 SINGLE-TARGET RESISTANCE
- 4 MANUFACTURING COSTS
- 5 ENVIRONMENTAL DEGRADATION
- 6 INEFFICIENT DELIVERY FORMULAS

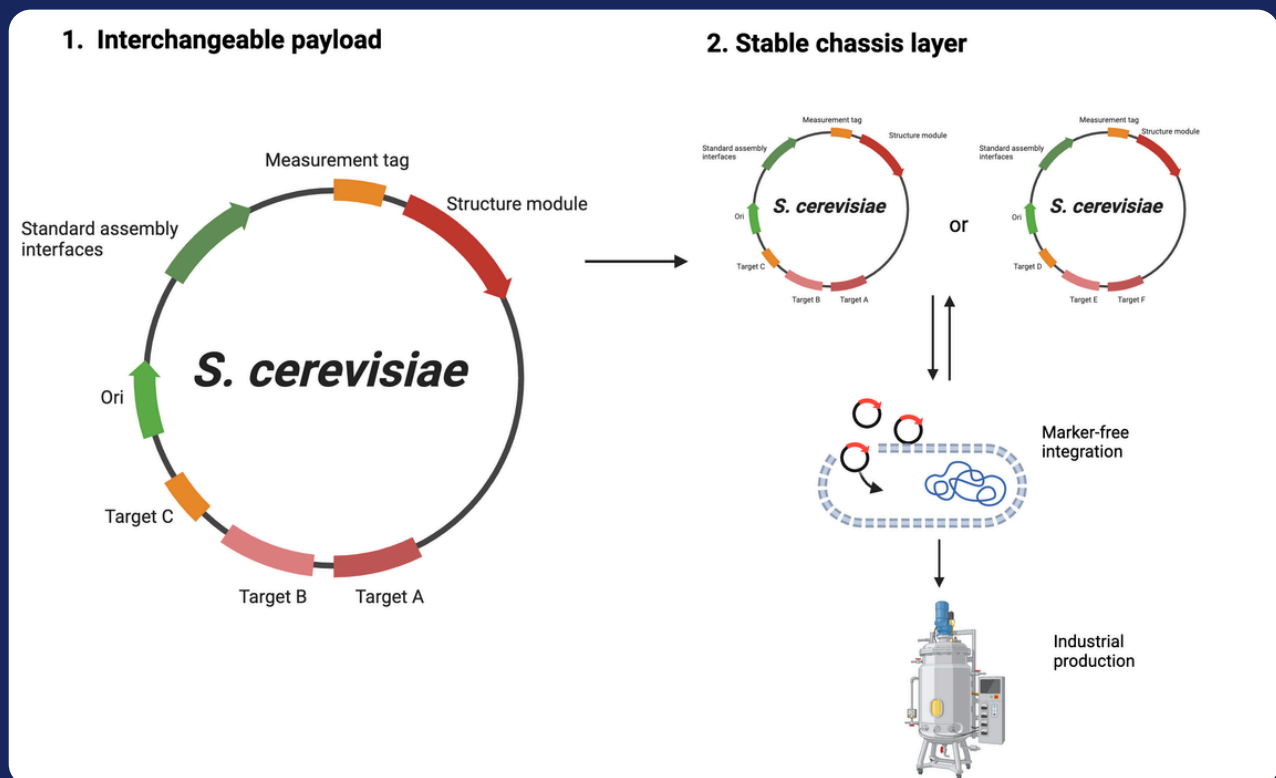
Our Solution

NECTAR is an **end-to-end, sequence-agnostic platform** connecting **computational design, modular genetic assembly, industrial yeast fermentation, and application-specific formulation**. It combines target design, transcript accessibility, and exhaustive siRNA off-target screening to identify potent, ecologically selective target regions.



Standardised cassettes enable multiplexed knockdown and screening of different RNA structures and RNA-binding modules for improved stability and localisation - a reusable design-build-test cycle (discover, de-risk, assemble, express, formulate, validate) in which experimental results feed back into each new design.

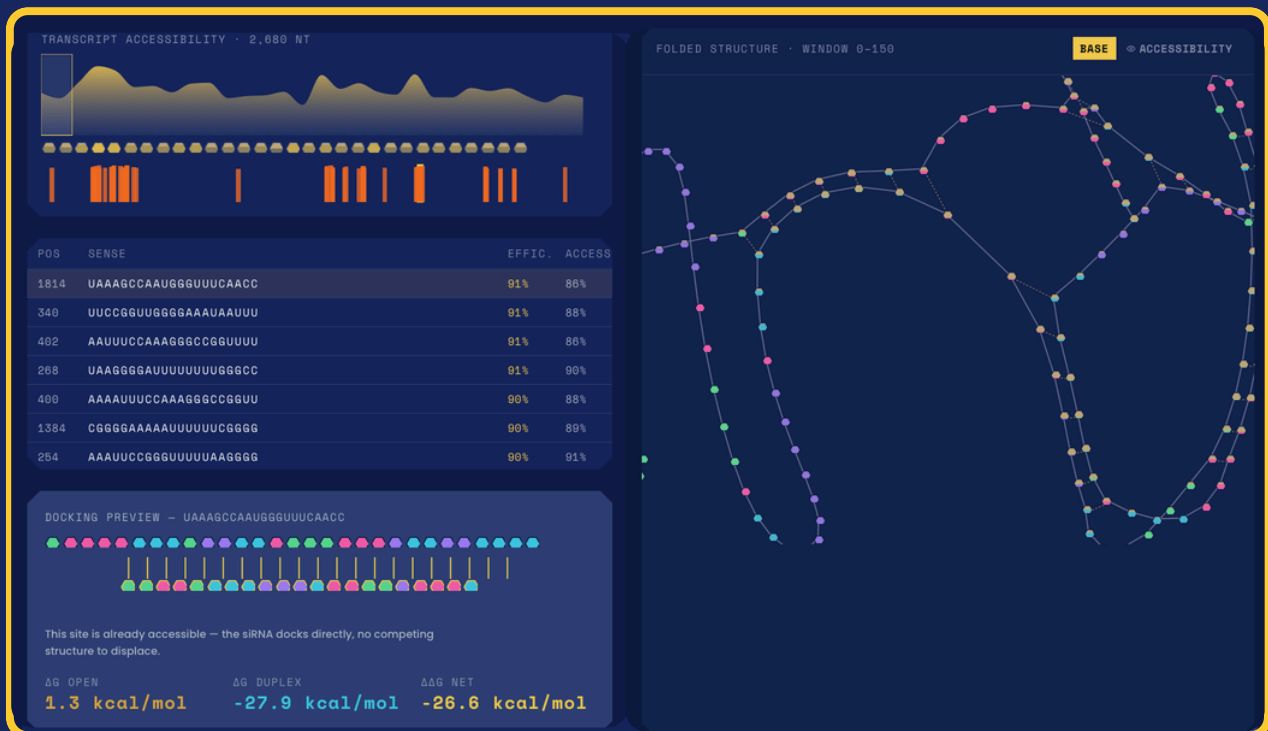
At the manufacturing level, engineered *Saccharomyces cerevisiae* becomes both a **factory and carrier**, increasing stability and deployability. Yeast lacks a canonical RNAi pathway, is readily engineered, and already benefits from mature industrial fermentation infrastructure. Rather than an additional RNA extraction step, cells can be harvested, inactivated and formulated directly. Our model estimates **5.2 tonnes of dried therapeutic yeast per 100 m³ batch at £3–8/kg**, with industrial therapeutic yeast fermentation taking advantage of existing infrastructure to scale production.



Modular engineering allows different NECTAR-designed targets to be integrated seamlessly into a yeast chassis for industrial production.

RNA design pipeline

accessible online:
NectarDesigner.bio



NECTAR's first use case focuses on applying **machine learning and AI-accelerated** systems to RNAi technologies to develop in-house predictive tools capable of designing optimised sequences for RNAi applications. The platform is trained on literature-derived data and subsequently validated through experiments conducted by our team, creating an end-to-end development pipeline.

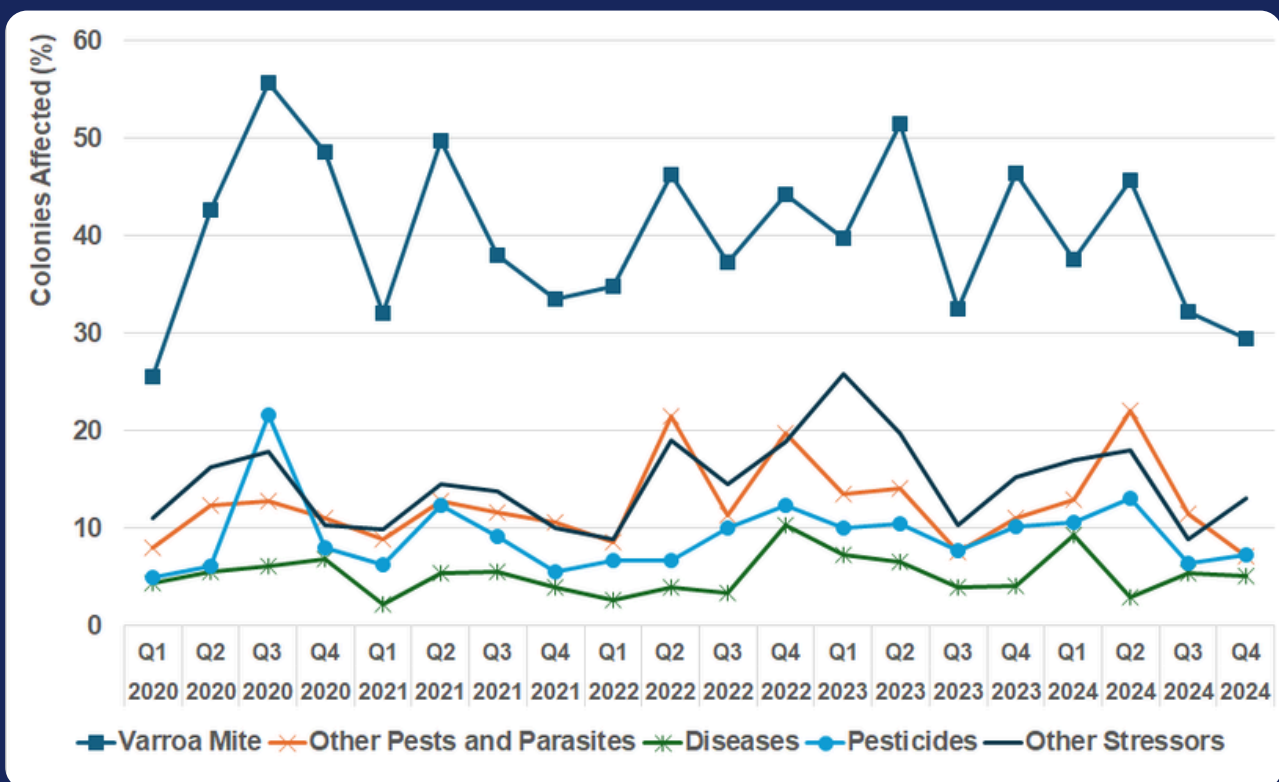
NECTAR enhances target design by:

- >**Modelling** accessible sequences in mRNA secondary structures
- >**Screening** for off-target homology within ecological context
- >**Optimising** construct design to create a **fully functional, expression ready cassette** which can be expressed in a range of vectors and chassis

Deployment case: *Varroa Destructor*

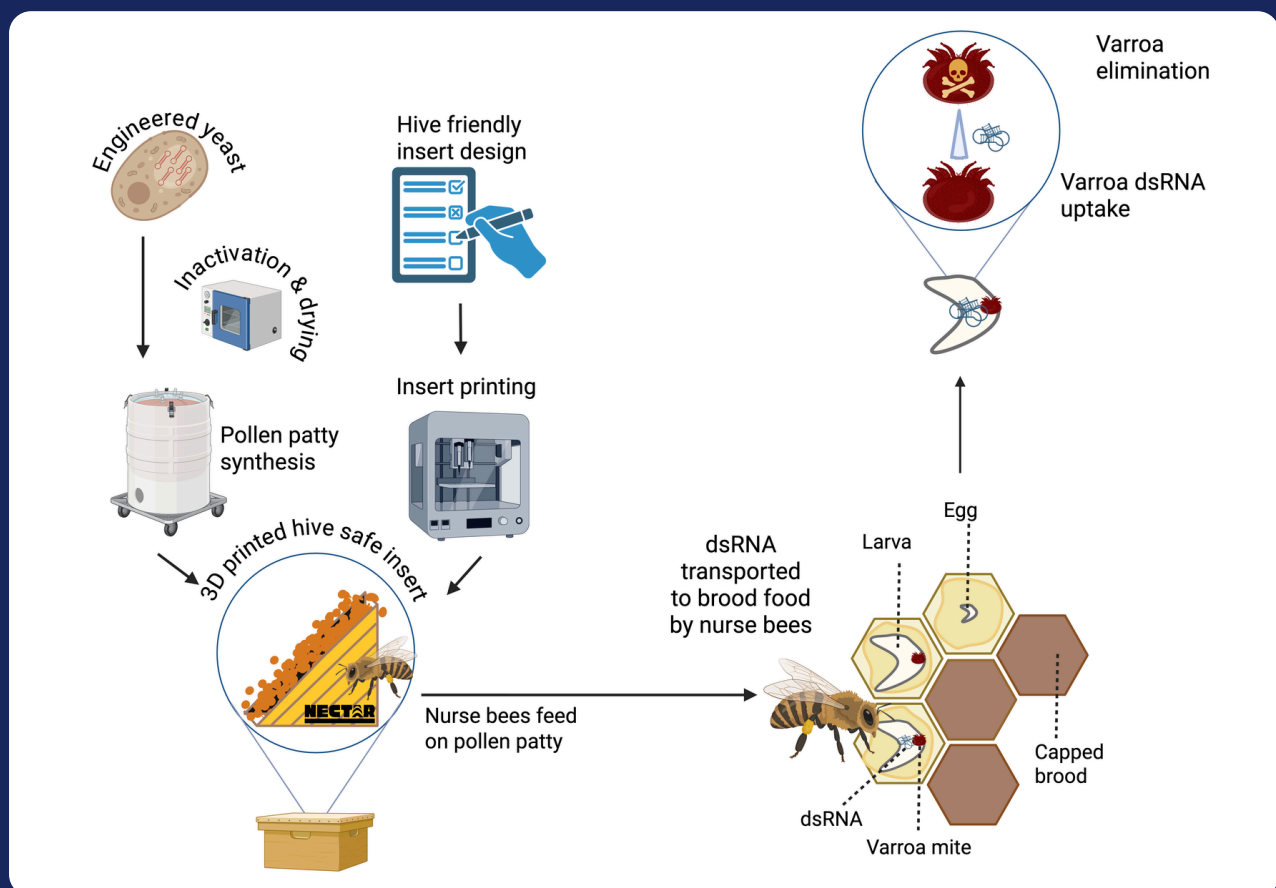
Varroa destructor, responsible for the majority of honey bee colony losses, is our first deployment case. In the US alone, honeybee-driven pollination adds \$18 billion to the agricultural economy, highlighting the need for a scalable, effective therapy to prevent the annual loss of more than half of the world's commercial honey bee operations.

Percentage of U.S Honey Bee Colonies Affected by Reported Stressors by Quarter, 2020-2024



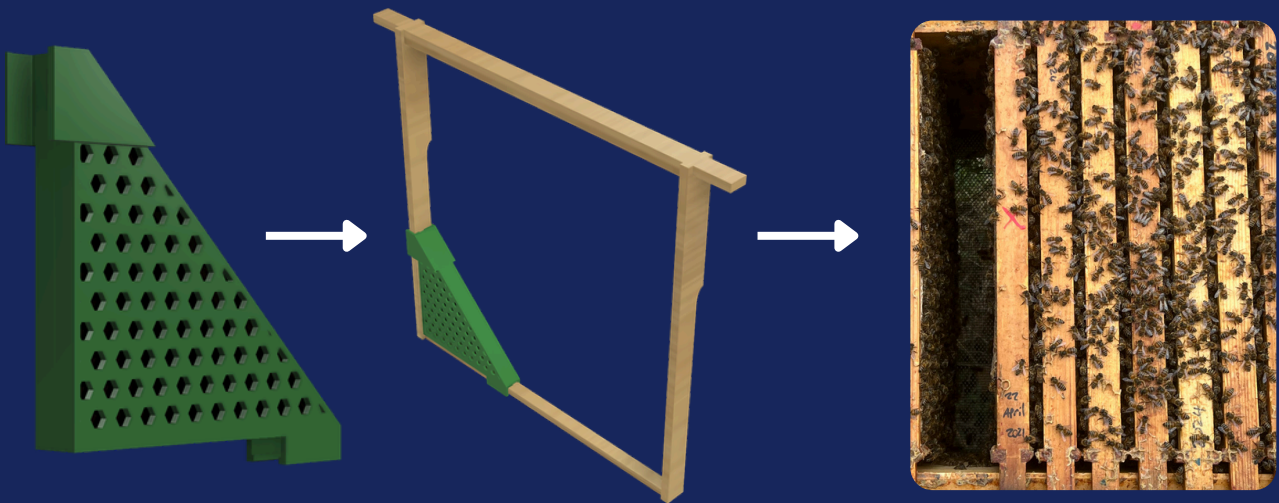
Existing dietary RNAi has reduced field infestation by 33–42%, proving feasibility while leaving scope to **improve potency, resistance management, and delivery.**

NECTAR designs a multi-target yeast formulation combining mite survival, reproduction and viral targets. Dried yeast is incorporated into a pollen patty and delivered through a 3D-printed hive insert.



One 100 m³ bioreactor alone could generate more than **five million 100 g treatment-equivalents** annually, reaching a gross sales capacity of over **\$100 million at \$20 per treatment**.

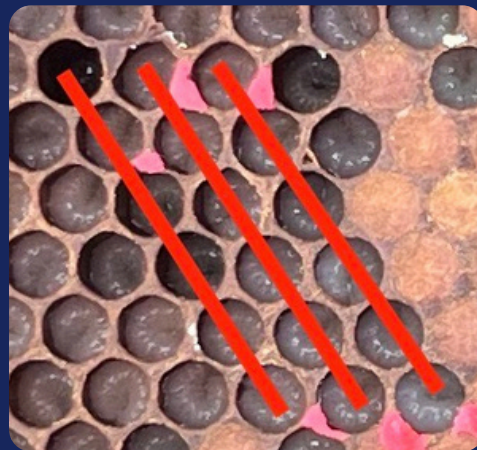
Our team has designed a 3Dprintable bio-safe insert that clips directly into hive frames, **minimising wasted space while maximising access to treatment** for both bees and beekeepers. The insert harnesses the natural transfer of RNA within the hive: nurse bees consume the therapeutic patty and deliver its dsRNA payload to developing larvae, where Varroa mites cause the greatest damage to the colony.



Each frame contains up to 250g of our NECTAR-formulated pollen patty, and is easily replaceable during routine hive monitoring.

Using NECTAR's RNA-design platform, we have identified 8 unique targets corresponding to essential mite reproductive and housekeeping genes. These targets were selected from thousands of candidates through mRNA secondary-structure analysis, assessment of interactions with native RNAi machinery, and screening for off-target sequence homology. To validate NECTAR-derived designs in vivo, we are using in vitro transcription-derived dsRNA to assess knockdown potency in mites, alongside stability and off-target toxicity in adult bees and larvae.

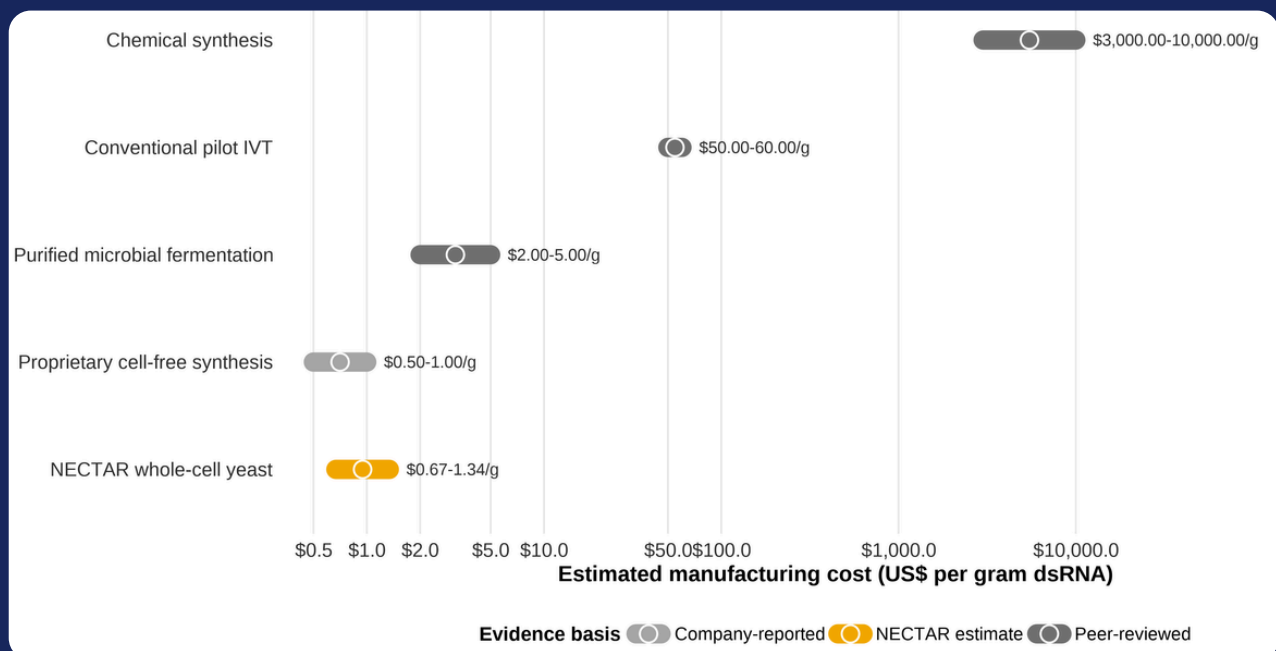
Adult and Larval feeding assays



We are working to further standardise dsRNA design and production by **comparing alternative dsRNA secondary structure architectures**. The results of these in vivo experiments will feed directly back into the continued refinement and validation of NECTAR's RNA-design platform.

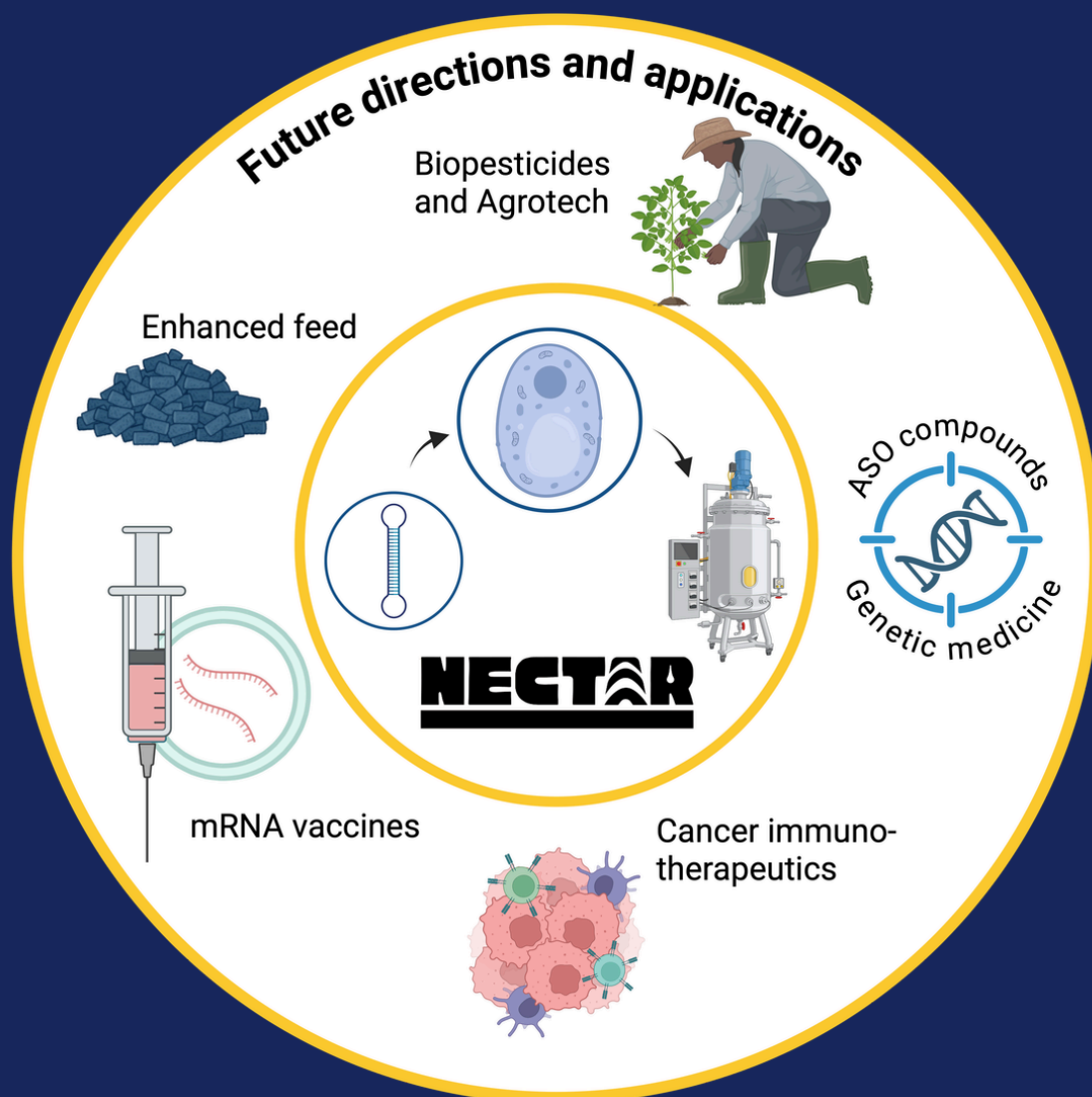
Commercialisation

Estimated manufacturing cost per gram of dsRNA active across production platforms. Bars distinguish peer-reviewed pilot estimates, company-reported commercial values and NECTAR's internal RNA-equivalent model. NECTAR costs refer to intact dsRNA contained within delivery-ready dried yeast and are not equivalent to purified products.



Conventional microbial and cell free production require fermentation, cell disruption, RNA extraction and purification before formulation. NECTAR instead uses engineered yeast as both the production chassis and protective carrier, allowing harvested and inactivated cells to be **incorporated directly into a treatment ready pollen-patty formulation.**

Applications



The applications of NECTAR span far beyond biopesticides. RNA technologies are currently in high demand in many fields of biotechnology. Critically, the value of NECTAR extends beyond one pesticide: it is a **programmable, purification-light, RNA biomanufacturing** platform combining safe design principles, resistance management, and industrial scalability to tackle a range of problems.

Team (all University of Oxford)

NECTAR

Team Mentor & PI

HAFEZ

Is only protecting the bees to keep the honey flowing



1st Year
Biology
St Catz

Wet Lab, Bee Lab, R&D

ANNEE

Fun fact: Opened for Boney M at John's Ball



2nd Year
Biology
St Catz

Wet Lab, Bee Lab, R&D

FREDDIE

Fav pipette size: p1000, most accurate for my motor skills



1st Year
Biology
Queens

Wet Lab, Bee Lab

MICHAEL

If you were a bee what bee would you be? Mosker bee cause I'm married to the grind



2nd Year
Biomedical Science
Queens

Wet Lab, Bee Lab, Dry Lab, Fundraising

THEODOR

Fav snack for long lab days: MILK
Number of tabs open: 274



1st Year
Biomedical Science
Lady Margaret Hall

Wet Lab, Bee Lab, PR

ARIANA

Fun fact: Has competed in both water polo and ice skating



1st Year
Biochemistry
St Hildas

Wet Lab, Bee Lab, Human Practices, PR

SAMMI

Fun fact: has been to the northernmost town in the world



2nd Year
Biology
St Catz

Wet Lab, Bee Lab, Human Practices

DIOGO

Hidden talent: Can (almost) juggle



2nd Year
Biology
Hertford

Wet Lab, Bee Lab, Fundraising

ROMAN

Fav tv show: breaking bad



1st Year
Biology
Lady Margaret Hall

Human Practices, Dry Lab

ALIYAN

Fun fact: Is the drummer in 2 bands and a percussionist for OUMS



1st Year
Engineering
Univ

Dry Lab, Human Practices

GRACIE

Hidden talent: Jumping across my 2 arms held together



2nd Year
Biochemistry
St Annes

Human Practices

ELLA

Hidden talent: capoeira and zumba (she started this month)



2nd Year
Biology
Brasenose

Wet Lab, Bee Lab, Human Practices

ANNA

Fav bee pun: What do you call an indecisive bee? a maybe



1st Year
Biomedical Science
New

Human Practices

ELIE

Fav scientist: Dorothy Hodgkin because she went to Somerville



1st Year
Biochemistry
Somerville

Human Practices

LISA

Fav song to pipette to: Another Day of Sun from La La Land



1st Year
Biochemistry
Queens

Human Practices

LAURA

Fav piece of lab equipment? A Buchner funnel because I work well under pressure



3rd Year
Chemistry
St Annes